[24] Recent Theoretical Advances in Non-Convex Optimization

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RQ2 / RQ3 / RQ4 — survey of generic complexity theory and structural assumptions.

Problem. A global minimum of a general non-convex objective can be computationally intractable to find. Which meaningful weaker targets have provable algorithms and complexity bounds? **Method.** Survey of deterministic, stochastic, higher-order, zeroth-order and structured non-convex optimization. **Main synthesis.** Guarantees depend on the target notion (stationarity or global optimality), assumptions and oracle access.

1. First-order stationarity versus global optimality

For $f : \mathbb{R}^d \to \mathbb{R}$ a standard algorithmic target is $\|\nabla f(x)\| \le \epsilon$. It can include saddles: for $f(x, y) = x^2 - y^2$, the origin has zero gradient but negative Hessian curvature.

If f has an L-Lipschitz gradient, gradient descent with step $1/L$ satisfies

$$f(x_{k+1}) \le f(x_k) - \frac{1}{2L}\|\nabla f(x_k)\|^2.$$

Summing from $k = 0$ to $T - 1$ gives

$$\frac{1}{T}\sum_{k=0}^{T-1}\|\nabla f(x_k)\|^2 \le \frac{2L[f(x_0) - f_*]}{T}.$$

Thus an iterate with gradient norm at most ϵ exists within $O(L\Delta/\epsilon^2)$ gradient evaluations. This is a *generic baseline*, not a deep-network global-optimality guarantee.

2. Stochastic gradient complexity requires a different metric

For g_k with $\mathbb{E}[g_k \mid x_k] = \nabla f(x_k)$ and bounded conditional variance $\mathbb{E}\|g_k - \nabla f(x_k)\|^2 \le \sigma^2$, a standard descent calculation yields

$$\frac{1}{T}\sum_{k=0}^{T-1}\mathbb{E}\|\nabla f(x_k)\|^2 \lesssim \frac{\Delta}{\eta T} + L\eta\sigma^2.$$

Choosing η on the order of $T^{-1/2}$ gives the familiar $O(T^{-1/2})$ squared-gradient measure and hence an $O(\epsilon^{-4})$ *stochastic-gradient baseline* for $\mathbb{E}\|\nabla f\|^2 \le \epsilon^2$. Variance reduction or stronger oracle assumptions may improve this rate.

3. Structural properties may permit much stronger claims

Under a Polyak–Łojasiewicz (PL) condition, for some $\mu > 0$,

$$\frac{1}{2}\|\nabla f(x)\|^2 \ge \mu\big(f(x) - f_*\big),$$

the same GD descent inequality implies $f(x_{k+1}) - f_* \le (1 - \mu/L)(f(x_k) - f_*)$ for appropriate constants. PL is weaker than convexity but excludes non-optimal stationary points; it must not be assumed for general neural networks.

4. Higher-order and zeroth-order methods

A second-order stationary target additionally requires $\lambda_{\min}(\nabla^2 f(x)) \gtrsim -\gamma$. Negative curvature can aid saddle escape, but Hessian/HVP queries and inner subproblem costs cannot be compared directly with a gradient-only oracle count. Zeroth-order methods estimate derivatives from function values and introduce additional dimension and noise dependencies.

Do not misattribute. The review organizes results from many independent papers. The GD and SGD calculations above are standard first-principles derivations included for teaching; they are not new theorems due to Danilova et al. A rate is meaningful only alongside its tolerance definition, regularity, and oracle model.

Survey connection. Best used to contextualize our Section V complexity table and Section VIII's assumption audit. **Our Notes.** Useful for preventing apples-to-oranges comparisons: two algorithms may have different complexity exponents because they solve different stationarity problems with different information.

Source: https://arxiv.org/abs/2012.06188